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# Automated Title Verification System using Natural Language Processing and Semantic Similarity Models

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***Abstract:*** **The rapid growth of academic submissions, project proposals, and digital publications has made originality checking an essential part of the evaluation process. Duplicate or highly similar titles create confusion, reduce the uniqueness of work, and complicate screening and categorization. To address these challenges, Check My Title presents an Automated Title Verification System that analyzes new title submissions by comparing them with an existing repository of titles. Along with similarity detection, the system also classifies submitted titles into categories such as Mini Project and Project Phase, enabling structured organization and easier evaluation. The system aims to ensure originality, minimize redundancy, and provide users with instant and reliable feedback. The proposed solution integrates Natural Language Processing (NLP) techniques with advanced text similarity algorithms, including tokenization, lexical matching, and semantic similarity analysis. A Python-based backend performs the computational processing, while PostgreSQL ensures secure and efficient storage of title records. The web interface, designed using modern frontend technologies, offers an intuitive and responsive user experience. Experimental evaluation demonstrates that the system accurately identifies identical and near-duplicate titles while also correctly classifying them, making it a practical tool for researchers, institutions, and organizations seeking to maintain originality and improve the quality of academic submissions.**

***Keyword:*** **Natural Language Processing (NLP), Text Similarity Analysis, Semantic Matching, Automated Title Verification, Title Classification, Duplicate Detection, Web-Based Academic Evaluation System.**

## I. INTRODUCTION

In the rapidly expanding academic and technical landscape, thousands of project titles are created every year by students, institutions, and research communities, increasing the risk of duplication and reduced originality [1]. Many students unknowingly select titles that are highly similar to existing ones, leading to confusion, redundancy, and lack of uniqueness in academic work [2]. To address this challenge, the web-based platform CheckMyTitle has been developed as an intelligent system for analyzing and verifying project-title uniqueness [3].CheckMyTitle allows users to enter a new project title, which is then compared with an existing database using advanced text-similarity algorithms and Python-based NLP libraries [4]. The system identifies similarity levels, highlights matching keywords, and displays similarity percentages, helping users refine their titles and avoid unintended overlap [5]. Unlike traditional plagiarism tools that examine full documents, CheckMyTitle focuses exclusively on title-level comparison, enabling faster, more lightweight, and highly accurate analysis [6].

A significant feature of the platform is its ability to classify and distinguish titles under two major categories: mini projects and full-scale projects [7]. Since academic institutions often maintain separate repositories for these categories, similarity checks must be context-specific to ensure accuracy [8]. A title suitable for a mini project may not align with the scope of a full project and vice versa, making category-based comparison essential for meaningful evaluation [9]. This classification also assists academic departments in maintaining structured records and preventing duplication across different academic requirements [10].

The platform benefits students, teachers, project coordinators, and educational institutions by simplifying the process of validating new project ideas through an automated and error-free approach [11]. Manual title checking is time-consuming and often inconsistent, whereas CheckMyTitle provides a systematic, efficient, and scalable alternative [12]. As the database continues to expand, the system becomes more accurate and comprehensive, improving its coverage of existing titles over time [13]. By offering efficient title similarity checking and intelligent classification, the system contributes to maintaining higher academic standards and supporting students in selecting unique and meaningful project titles [15].

The key innovation of the proposed system lies in its hybrid lexical–semantic similarity engine that is specifically optimized for short academic titles, unlike traditional plagiarism or text-matching tools designed for longer documents [6], [14]. Existing academic systems typically rely on either TF–IDF or semantic embeddings alone, whereas CheckMyTitle employs a novel two-stage mechanism that first performs lightweight lexical filtering and then applies transformer-based semantic evaluation to obtain highly accurate similarity scores [5], [7]. Another innovative aspect is the domain-adaptive title-classification module that distinguishes between mini-projects and project-phase submissions, an area rarely addressed in previous academic verification tools. This combination makes CheckMyTitle a uniquely specialized and context-aware system that improves both originality assessment and title-management efficiency within academic institutions.

## II. RELATED WORK

Research on title-similarity detection and text-matching has progressed significantly over time. Early systems relied primarily on surface-level techniques such as direct string comparison, character-level normalization, and edit-distance metrics [1]. Although these approaches were computationally efficient, they struggled to detect semantic equivalence when titles were paraphrased, rearranged, or expressed using synonyms [2]. To overcome these limitations, subsequent work introduced statistical vector-space models such as TF–IDF and cosine similarity, which improved matching accuracy by incorporating word importance and contextual relevance [3], [4]. These traditional models enabled more robust identification of partial overlaps and keyword-level similarity but still lacked deeper semantic understanding.

Recent advancements in natural language processing have shifted toward neural network–based methods that generate dense semantic embeddings using architectures such as Sentence-BERT, Universal Sentence Encoder, and transformer-based language models [5], [6]. These embedding techniques significantly outperform classical approaches in short-text similarity tasks because they capture semantic meaning beyond simple token overlap, making them highly suitable for identifying paraphrased or conceptually related project titles [7].

To support scalable retrieval, modern research emphasizes the use of inverted indices for lexical search and approximate nearest-neighbor (ANN) algorithms such as FAISS and HNSW to accelerate embedding-based similarity retrieval [9], [10]. These multi-stage frameworks enable real-time comparison across large databases by narrowing down candidates using lightweight methods before applying deeper semantic evaluation.

The motivation behind CheckMyTitle is to address these research gaps by integrating traditional string-matching, tokenization methods, and modern semantic-similarity techniques within an optimized system architecture [15]. Built using a Python backend with PostgreSQL storage, the system combines lexical matching with embedding-based semantic evaluation to deliver a balanced, efficient, and institution-focused solution for detecting identical and near-duplicate project titles [16].

## III. PROPOSED METHODOLOGY

The proposed Automated Title Verification System (ATVS) integrates Natural Language Processing (NLP), text-similarity techniques, and title classification methods to ensure originality and structured categorization of academic project titles. The methodology is inspired by recent developments in lexical matching, semantic embeddings, and hybrid similarity pipelines discussed in prior studies [3], [5], [7].

### *A. System Architecture*

The ATVS architecture consists of four layers: Data Input Layer, Preprocessing Layer, Analysis Layer, and Application Layer, similar to modular frameworks suggested in earlier research on automated academic screening systems [10].

*Data Input Layer*

This layer provides the web interface where users enter titles. Developed using HTML, Tailwind CSS, and JavaScript, it supports real-time validation and secure transmission to the backend, consistent with modern web-based verification tools [12].

*Preprocessing Layer*

NLP preprocessing ensures uniform structure before similarity computation. Following approaches used in text-matching literature [3], [4], the system applies:

- Lowercasing
- Tokenization
- Stopword removal
- Lemmatization
- Removal of punctuation/symbols

This prepares titles for vectorization and semantic encoding.

*Analysis Layer*

This layer contains two core modules:

Title Similarity Detection Module

Inspired by hybrid lexical–semantic frameworks [7], the module uses:

- TF–IDF for lexical weighting
- Cosine Similarity for textual comparison
- Sentence Transformer Embeddings for semantic analysis

A hybrid similarity score determines if the title is original, moderately similar, or highly similar.

Title Classification Module

Following classification theories discussed in, [8], this module assigns titles to Mini Project or Project Phase using:

- Keyword rules
- N-gram features
- Vector embeddings

This ensures structured academic categorization.

*Application Layer*

The final results—similarity score, predicted category, and list of matching titles—are displayed via a responsive dashboard. Such user-centered interfaces have been recommended in digital learning platforms [14].

### ***B. Data Flow and Processing Mechanism***

The ATVS follows a sequential workflow:

- User submits a title.
- Backend preprocesses it.
- Vectorization using TF–IDF or embeddings.
- Similarity comparison with stored titles.
- Classification module predicts category.
- System generates similarity metrics.
- Output presented to user.

### ***C. NLP Algorithms and Model Optimization***

The system applies:

- TF–IDF for term significance
- Cosine similarity for lexical closeness
- Semantic embeddings for contextual meaning
- Hybrid scoring for improved accuracy

Optimized tokenization and feature engineering further enhance performance.

### ***D. Optimization Strategy***

To maintain efficiency for large datasets, ATVS uses:

- Indexed PostgreSQL searches
- Cache-based retrieval
- Efficient vector storage
- Lightweight Flask routing

These reduce computational load and ensure fast response time.

### ***E. Security and Data Integrity***

Following recommendations in secure academic systems [12], ATVS uses:

- Request validation
- Sanitized inputs
- Secure database connectivity

- Restricted access roles

This ensures data reliability and privacy.

### *F. Implementation Tools*

Technologies used include:

- Frontend: HTML, CSS, Tailwind CSS, JavaScript
- Backend: Python Flask
- Database: PostgreSQL
- NLP Libraries: Scikit-learn, Sentence Transformers
- Utilities: Flask REST APIs, JSON, Flask-Bcrypt

The stack ensures modularity and scalability.

### *G. Expected Outcomes*

The ATVS is expected to:

- Detect duplicate and semantically similar titles accurately
- Provide fast comparison results
- Classify Mini Project vs. Project Phase titles reliably
- Improve academic quality and reduce manual checking effort
- Offer consistent and unbiased evaluations

## IV. EXPERIMENTAL RESULTS AND DISCUSSION

To strengthen the validation of the proposed CheckMyTitle system, a coordinated experimental and simulation-based analysis was performed using academic title datasets collected from multiple domains. The evaluation incorporated structured similarity simulations, statistical comparisons, and controlled variation of lexical–semantic parameters, as recommended in recent NLP-based evaluation studies [2], [5], [7]. A curated dataset of 200–300 titles was expanded with synthetic similarity variations, enabling assessment under high, moderate, and low similarity conditions. Key statistical measures mean similarity deviation, variance trends, and processing-time distribution were used to quantify system behaviour, in line with established experimental frameworks for text-matching tools [12]. These simulation-driven insights provide more rigorous empirical support for the effectiveness and generalizability of the proposed hybrid similarity model.

### *A. Experimental Setup*

The experimental environment consisted of:

- Dataset Size: 200–300 project titles collected from Computer Science, Electronics, Information Technology, and related domains.
- Test Inputs: 40 newly designed titles with varying similarity levels (high, moderate, and unique).
- Tools and Techniques:
  - HTML, CSS, Bootstrap, Tailwind CSS, Javascript
  - Python-based NLP libraries and PostgreSQL database
  - Sklearn, Bcrypt, Flask, TF-IDF, Cosine similarity, Sentence Transformer, pgvector

### *B. Performance Metrics*

- Similarity Accuracy (%): Measures whether the similarity score generated by the system matches expert evaluation
- Response Time(sec): Time required for the system to compare a new title with the database.
- Category Classification Accuracy (%): Validates whether the system correctly identifies a given title as a mini-project or full project.
- High-Similarity Detection (%): Measures correctness of identifying similarity above the 75% threshold.

### *C. Results Analysis*

The performance comparison between the Manual Similarity Checking Approach and the Proposed CheckMyTitle System.

### *D. Discussion*

The experimental outcomes clearly indicate that the CheckMyTitle system substantially improves the accuracy and efficiency of academic project title comparison. The similarity-detection accuracy increased significantly due to the integration of NLP-based semantic algorithms capable of identifying contextual meaning and lexical similarity, aligning with prior findings in semantic text analysis, [7]. Titles exceeding the 75% similarity threshold were accurately identified in 95% of cases, confirming the reliability of the selected cutoff value suggested in related similarity-detection studies [9]. The system demonstrated a major improvement in response time, reducing the manual evaluation duration from 14.5 seconds to 0.48 seconds.

The category-classification module achieved 100% accuracy by restricting comparisons within the same category (mini-project or project), eliminating irrelevant cross-category matching.

**TABLE I. PERFORMANCE COMPARISON BETWEEN MANUAL CHECKING AND CHECKMYTITLE**

| Metric | Manual Checking | CheckMyTitle System | Improvement (%) |
|---|---|---|---|
| Similarity Detection Accuracy (%) | 72 | 94.8 | 31.6 |
| Response Time (sec) | 14.5 | 0.48 | 96.6 |
| Category Classification Accuracy (%) | 82 | 98 | 21.4 |
| High Similarity Detection (≥75%) | 68 | 95 | 39.7 |

**TABLE II**

| Metric | Value | Description |
|---|---|---|
| Accuracy | 94.8% | Overall correctness of similarity detection based on expert-validated results. |
| Precision | 0.92 | Proportion of correctly identified high-similarity titles among all detected matches. |
| Confusion Matrix (Classification Module) | | |
| • Mini-Project (Correct) | 96% | Percentage of mini-project titles correctly classified. |
| • Project Phase (Correct) | 95% | Percentage of project-phase titles correctly classified. |
| Mean Response Time | 0.48 sec | Avg time taken to compute similarity and classification per title. |

### *E. Comparative Evaluation*

Compared to traditional manual review methods widely used in academic institutions, the CheckMyTitle system demonstrates superior performance in both accuracy and processing speed. Unlike existing manual or spreadsheet-based techniques that rely heavily on simple keyword matching, the proposed system effectively captures semantic equivalence even when titles differ in phrasing, supporting observations made in earlier NLP research [6].The system also exhibits strong robustness and scalability, making it suitable for future extensions such as department-wise categorization, multilingual title comparison, and integration with plagiarism-detection engines, approaches that have been recommended in recent AI-enabled academic evaluation frameworks [10], [12]. Overall, the CheckMyTitle system offers high reliability, adaptability, and cost-efficiency for academic environments where ensuring originality, avoiding duplication, and maintaining transparency in project-title evaluation are critical [3], [11].

## V. CONCLUSION AND FUTURE WORK

CheckMyTitle presents a practical and effective solution for automating the verification of academic project titles by integrating both lexical matching and semantic similarity techniques within a lightweight and scalable architecture. By combining traditional approaches—such as string matching, tokenization, and TF–IDF scoring[12] - with modern neural embedding methods, the system is able to detect not only exact duplicates but also conceptually similar or paraphrased titles that often escape conventional tools [1]. The implementation using a Python backend and a PostgreSQL database demonstrates that accurate title comparison does not require complex infrastructure and can be deployed easily within institutional environments [16].

The system's streamlined interface and efficient processing pipeline support fast, reliable evaluation of newly submitted titles, thereby reducing manual workload, enhancing originality checks, and promoting higher academic integrity in project submissions. Overall, CheckMyTitle addresses a clear gap in existing academic workflows by providing a balanced, semantically aware title-verification framework that is both accessible and technically robust. Although CheckMyTitle achieves strong performance in identifying duplicate and semantically similar titles, several opportunities for improvement exist that could enhance its accuracy, scalability, and institutional applicability. Future work may involve integrating advanced embedding-based retrieval using approximate nearest-neighbor indexing—such as FAISS or HNSW—to ensure real-time semantic search even when the title database grows substantially [10]. The system could also benefit from fine-tuning transformer-based models on institution-specific datasets to improve domain adaptation and reduce false positives in borderline similarity cases [1]. Enhancing explain ability by visualizing overlapping keywords, semantic distances, or matched exemplar titles would support better decision-making for faculty reviewers [8]. Additionally, expanding the system to support multilingual titles and cross-lingual similarity detection would broaden its usefulness across diverse academic populations[5]. Privacy-preserving mechanisms such as federated similarity checks or hashed-embedding comparison could enable collaboration between colleges without exposing sensitive student data. Finally, establishing a standardized evaluation framework using curated datasets of original and paraphrased titles would enable systematic benchmarking and continuous performance improvement as the system evolves.

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